MIL-T-45088B(Ord)
9 March 1962

SUPERSEDING
MIL-T-45088A(Ord)
20 May 1960

MILITARY SPECIFICATION

TRANSMITTER, SUPERELEVATION: M23

1. SCOPE

1.1 This specification covers one type of electro-mechanical instrument utilized to subtract gun elevation data from superelevation data supplied from a computer and transmit the difference electrically to an elevation servomechanism.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock Corner
PPP-B-636	Boxes; Fiber
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-60	Tape; Pressure-Sensitive Adhesive, Waterproof For Packaging and Sealing

FSC 1265

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-P-130	Paper, Wrapping, Laminated and Creped
MIL-B-131	Barrier Material, Waterproof, Flexible
MIL-C-3133	Cellular Rubber, Products, General Purpose (Except Ebonite)
MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-A-5092	Adhesive Rubber (Synthetic and Reclaimed Rubber Base)
MIL-R-6855	Rubber; Synthetic, Sheet, Molded, and Extruded, for Aircraft Applications
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-S-20708	Synchros, 60 and 400 cycles, General Specification
MIL-I-45208	Inspection Requirements, General Specifications for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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DRAWINGS

Ordnance Corps

F8298984	Transmitter, Superelevation: M23
B8298954	Wiring Diagram - Transmitter, Superelevation: M23
LSQAP-8298984	List of Supplementary Quality Assurance Provisions for Transmitter, Superelevation: M23

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one transmitter representative of production, shall be made available to the Government to determine if the transmitter meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The transmitter shall be manufactured in accordance with Drawing F8298984, "Transmitter, Superelevation: M23" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

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3.5 Environmental.

3.5.1 Storage temperatures.- The transmitter shall meet all requirements of this specification at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.5.2 Operating temperatures.- The transmitter shall meet the applicable requirements of 3.8 while exposed to and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

3.5.3 Vibration.- With the transmitter mounted and secured in its normal operating position (the arm at 6 degrees elevation with reference to the base) and with the Cam D8298974 immobilized, the transmitter shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no loose or missing parts and the transmitter shall meet all requirements of this specification

3.6 Electrical circuits.

3.6.1 Continuity.- The wiring of the transmitter shall demonstrate continuity in accordance with Wiring Diagram B8298954.

3.6.2 High potential.- Insulation breakdown shall not occur between isolated circuits and between ungrounded electrical circuits and the body or frame when 500 volts root mean square (rms) are applied for a period of at least 30 seconds.

3.6.3 Insulation resistance.- Following the high potential test, insulation resistance between ungrounded electrical circuits and the body or frame and between isolated circuits shall not be less than 2 megohms, with 500 volts direct current (dc) applied.

3.7 Accuracy.

3.7.1 Control differential synchros.- The coarse and fine differential synchros shall be at electrical synchro zero, when checked to the procedures of Specification MIL-S-20708/93 and the angular position of the arm is six degrees in accordance with Drawing F8298984.

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3.7.2 Superelevation.- Starting with the synchros positioned in accordance with 3.7.1, the shaft of the fine differential synchro shall rotate through an angle 128 times the angle generated by the movement of the input arm within 24 minutes, as measured by external electrical means, Readings shall be obtained at each extreme limit of movement and at three intermediate points. Readings shall be obtained by approaching each point from different directions of arm movement.

3.7.2.1 Coarse synchro.- By utilizing external electrical means, operation of the coarse control differential synchro shall be evaluated by determining the angular displacement of the rotor. When the arm is displaced through an angle of 6 degrees, the coarse synchro rotor shall rotate 15 degrees within 45 minutes.

3.8 Operability.- The maximum force required to elevate and depress the input arm through its complete cycle of operation, applying the force at the flat surface of the outboard bearing, shall not exceed 16 ounces at standard ambient temperature and at plus 150 degrees Fahrenheit, and shall not exceed the value obtained at standard ambient temperature by more than 25 ounces when subjected to extreme low temperature of 3.5.2.

3.9 Sealing.- The transmitter shall show no evidence of leakage in excess of 0.1 pound per square inch gage pressure (psig) when subjected to an internal pressure of 5 psig for a minimum period of 5 minutes. The transmitter shall not be immersed in water to determine any area of excess leakage.

3.10 Identification marking.- Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General requirements.- Inspection shall be in accordance with General Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one transmitter shall be selected by the Government inspector as soon as practicable after the start of production and shall be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8298984.

4.4.1 Contents of SQAP.- The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

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4.4.1.1 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the transmitter.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Accuracy requirements.- Inspection equipment which incprporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.1 Inspection equipment design requirements.

4.6.1.1 Device to test environmental requirements.- The device shall have means for controlling and recording the temperature conditions required and shall be such that the equipment under test shall not exceed fifty (50) percent of the internal volume of the test chamber and no portion shall be closer than six (6) inches from any surface of the chamber.

4.6.1.2 Device to test vibration requirements.- The device shall be of a type of vibrating machine capable of meeting the conditions required. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude and frequency within the allowable tolerance specified in Specification MIL-F-13926. The vibrating machine shall also be equipped with a medium capable of holding the transmitter securely in the normal operating position as depicted on Drawing F8298984, with the Arm Assembly F8298983 completely immobilized. The base of the medium shall be secured firmly to the vibrating machine.

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4.6.1.3 Device to test control differential synchros requirements.- Accuracies for all testing equipment required for acceptance inspection shall be within the limits outlined in Specification MIL-S-20708.

4.6.1.4 Device to test superelevation and coarse synchro requirements.- The device shall consist of a special test fixture capable of holding the Superelevation Transmitter: M23 in the normal operating position as shown on Ordnance Drawing F8298984.

4.6.1.4.1 Provisions shall be made to permit angular positioning of the input arm to any angular position specified in 3.7.2 within an accuracy of three (3) minutes.

4.6.1.4.2 The fixture shall be equipped with external electrical means capable of indicating and measuring the angular displacement of both the fine and coarse synchro transmitter rotor shafts.

4.6.1.4.3 The medium utilized for measuring the angular position of the fine synchro transmitter rotor shaft shall be accurate within three (3) minutes.

4.6.1.4.4 The medium utilized for measuring the angular position of the coarse synchro transmitter rotor shaft shall be accurate within five (5) minutes.

4.6.1.5 Device to test sealing requirement.- The device shall be capable of charging the interior of the Superelevation Transmitter: M23 to an internal pressure of 5 psig plus or minus .1 psig.

4.6.1.5.1 The device shall be equipped with a pressure gage capable of recording the internal pressures up to at least 5 psig subdivided into .1 psig or into increments whereby .1 psig may be readily estimated. The pressure gage shall be accurate to within 1/4 of 1 percent plus or minus of the maximum dial reading over the entire range.

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4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables I, II, III and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.0%	Req't Par.	QA. Par.	Test Method
101. Continuity	3.6.1	4.8.3.1	Standard Measuring Equipment (SME)
102. High potential	3.6.2	4.8.3.2	SME
103. Insulation resistance	3.6.3	4.8.3.3	SME
104. Synchro zero	3.7.1	4.6.1.3	MIL-S-20708
105. Rotation of fine synchro shaft	3.7.2	4.8.4	Device, Special Testing (DST)
106. Rotation of coarse synchro shaft	3.7.2.1	4.8.4	DST
107. Operability w/o subject to temperature test	3.8	4.8.5	DST SME-Tactile
108. Sealing	3.9	4.8.6	SME
109. Completeness of assy.	3.3	Per Dwg	Visual
110. Identification, marking	3.10	4.8.7	Visual

MINOR: NONE DEFINED

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table V of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Vibration	3.5.3	4.8.2.3	Device, Special Testing (DST)
102. Temperature, storage	3.5.1	4.8.2.1	SME-Tactile
103. Temperature, operating	3.5.2	4.8.2.2	SME-Tactile
104. Operability	3.8	4.8.4	SME-Tactile

MINOR: NONE DEFINED

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	Test Method
101. Improper material	5.1.1	Applicable Specification
102. Cleaning (Level A,C)	5.3.1.1 & 5.3.2.1	MIL-P-116
103. Drying (Level A,C)	5.3.1.2 & 5.3.2.1	MIL-P-116
104. Improper preservation (Level A)	5.3.1.3	MIL-P-116
105. Improper unit package (Level A)	5.3.1.4	Fig. 1 thru 3 Incl. & MIL-P-116

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MAJOR: AQL 1.5% (Cont'd)	Req't Par.	Test Method
106. Improper packaging of boot and clamp (Level A,C)	5.3.1.4.4 5.3.2.2	Visual
107. Immobilization of package on base (Level A,C)	5.3.1.4.6	Fig. 1 & 5- Visual
108. Improper packing (Level A)	5.4.1	Visual
109. Improper strapping (Level A,B,C)	5.4.3	Visual
110. Improper marking (Level A)	5.5.1 5.5.3	Visual
111. Improper marking (Level A,B)	5.5.2	Visual
112. Workmanship	5.7	Visual

MINOR: AQL 4.0%	Req't Par.	Test Method
201. Improper wrapping (Level A,C)	5.3.1.4.1 5.3.2.2	Visual
202. Improper wrapping of hardware (Level A,C)	5.3.1.4.2 5.3.2.2	Visual
203. Improper wrapping of pins and springs and mounting hardware (Level A,C)	5.3.1.4.3 5.3.2.2	Visual
204. Improper base (Level A,C)	5.3.1.4.5 5.3.2.2	Visual
205. Improper bag size	5.2	Visual
206. Improper packing (Level B,C)	5.4.2	Visual

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TABLE IV - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table V of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	Method of Protection
101. Heat seal (Level A)	5.3.1.4.7	MIL-P-116
102. Quick leak (Level A)	5.3.1.4.7	MIL-P-116

MINOR: NONE DEFINED

4.8 Test methods and procedures.

4.8.1 All performance tests conducted to determine compliance with the requirements specified herein shall be performed on the superelevation transmitter prior to complete assembly. Testing shall be performed prior to the assembly of Clamp 502942, Clamp Assembly 8205105, Boot 8298976, Springs 8298992 and Cam 8298974 depicted on Drawing F8298984 to the unit.

4.8.2 Environmental.

4.8.2.1 Storage temperature.- The transmitter shall be exposed to the temperatures indicated in 3.5.1 by means of test chambers capable of maintaining these temperatures. The test shall be conducted in accordance with Procedure II of Specification MIL-F-13926 to the specified conditions of 3.5.1. After exposure, the transmitter shall be subjected to and shall meet the storage temperature requirement.

4.8.2.2 Operating temperatures.- The transmitter shall be placed in a test chamber capable of maintaining the temperatures indicated in 3.5.2 and the test shall be conducted in accordance with Procedure I of Specification MIL-F-13926. The transmitter shall be subjected to and shall meet the requirements specified in 3.5.2.

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4.8.2.3 Vibration.- This test shall be the first test performed on the transmitter. The transmitter shall be vibrated as indicated in 3.5.3 by means of a special testing device with a medium to support the unit under test in the specified position. The input arm shall remain immobile throughout the test. At the conclusion of the vibration, the transmitter shall be subjected to a visual and tactile examination for any loose, missing or damaged parts. If there is no evidence of loose, missing or damaged parts, then the unit shall be tested for and shall meet the requirements of this specification.

4.8.3 Electrical.

4.8.3.1 Continuity.- A continuity check of the transmitter wiring shall be made using standard measuring equipment to determine conformance with the requirements of 3.6.1.

4.8.3.2 High potential.- The transmitter shall be tested in accordance with Method 301 of Standard MIL-STD-202, except that the duration of application of the test voltage specified in 3.6.2 shall apply to determine conformance with the requirements specified.

4.8.3.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for high potential, the transmitter shall be given the insulation resistance test in accordance with Method 302, test condition B, of Standard MIL-STD-202 to determine conformance with the requirements of 3.6.3.

4.8.4 Accuracy.- With the superelevation transmitter properly positioned on the special testing device and with the input arm set to six (6) degrees by the means provided on the special testing device, the unit shall indicate synchro zero setting on the test equipment to determine compliance with the requirements of 3.7.1. Reset input arm one (1) degree from the established position and then record reading observed on the special testing device to determine conformance with the requirements of 3.7.2. Reset arm six (6) degrees from the previous established position, the reading observed on the special testing device shall be within the limit specified in 3.7.2.1. Then reset input arm by means provided on the special testing device to maximum elevation position and maximum depression position. After each setting, record the readings observed on the special testing device. These readings shall be within the limits specified in 3.7.2. Select any three intermediate positions and the readings obtained on the special testing device at each position shall be within the limits specified in 3.7.2.

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4.8.5 Operability.- This test consists of measuring the force required to elevate and depress the input arm through its full travel while at standard ambient temperature and extreme temperatures. Readings shall be taken when the input arm is in motion using standard measuring equipment. The readings obtained while at standard ambient temperature shall not exceed the value specified in 3.8 and they shall be noted. During the performance testing at plus 150 degrees and minus 40 degrees Fahrenheit, the input arm shall again be motivated as above, and the readings obtained shall not exceed the values indicated in 3.8.

4.8.6 Sealing.- The test shall be accomplished by means of commercial equipment which shall include a sensitive pressure gage capable of testing for the sealing requirements. A pressure testing fitting shall be attached to Valve Assembly B8201751 in the transmitter housing provided for this purpose. The unit under test shall be subjected to an internal pressure indicated in 3.9 of dry nitrogen. Leakage, as indicated on the gage of the test equipment, shall not be evident within the time interval specified in 3.9.

4.8.7 Identification marking.- Marking for identification shall be visually checked for conformance with the requirements of 3.10.

4.8.8 Preproduction pack.- The preproduction pack shall be inspected by the contractor and witnessed by the Government representative to determine conformance with the requirements of 5.6.

4.8.9 Rough handling test.- The preproduction pack only shall be subjected to the rough handling test in accordance with the procedures outlined in Specification MIL-P-116.

4.8.10 Preproduction pack failure.- Should the preproduction pack fail to pass the acceptance tests, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be selected and completely inspected by the contractor and witnessed by the Government representative.

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4.8.11 Packaging tests.- Each package in the applicable sample shall be subjected to the pertinent packaging test of Specification MIL-P-116 as determined by the method of protection shown in Table IV.

4.8.12 Determination of cleanliness test.- Periodically, the applicable surfaces of one or more units shall be subjected to the "Determination of Cleanliness" test in accordance with Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

5.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Greaseproofed barrier material	Type I or II, grade A, class 2 of Specification MIL-B-121
Gummed tape	Class 2 of Specification UU-T-111
Paperboard cushioning material	Type I, class 1 of Specification PPP-P-291
Pressure sensitive tape	Type I or II of Specification UU-T-106
Waterproof pressure sensitive tape	Type 3, class 2 or 3 of Specification PPP-T-60
Water-vaporproof barrier	Class 1 of Specification MIL-B-131

5.2 Size of containers.- Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

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5.3 Preservation and packaging.

5.3.1 Level A.

5.3.1.1 Cleaning.- Cleaning Process C-1 of Specification MIL-P-116 shall be followed to clean all exterior surfaces of the transmitter and loose items.

5.3.1.2 Drying.- Immediately after cleaning, the surfaces shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

5.3.1.3 Preservative application.- All unprotected metal surfaces, including mounting hardware and locating pins, shall be coated with preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116). Preservative shall be applied on the unprotected metal surfaces by the brushing procedure specified in Specification MIL-P-116. Preservative shall be applied on the pins and hardware by any applicable procedure specified in Specification MIL-P-116. Excess preservative shall be removed.

5.3.1.4 Unit packaging.- The transmitter shall be packaged as specified herein and illustrated in Figures 1,2,3, 4 and 5, which is in conformance with Method IA-16 of Specification MIL-P-116.

5.3.1.4.1 Wrapping end of arm containing cam follower.- The end of the arm, which contains the cam follower, shall be wrapped with greaseproofed barrier material. The wrap shall be secured in place with pressure-sensitive tape.

5.3.1.4.2 Transmitter mounting hardware.- The mounting hardware shall be wrapped with greaseproofed barrier material and overwrapped with paperboard cushioning material. The wrap shall be secured in place with gummed tape. The wrapped hardware shall be secured to the arm of the transmitter with waterproof pressure-sensitive tape.

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5.3.1.4.3 Cam locating pins, springs and mounting hardware.- The locating pins shall be wrapped with greaseproofed barrier material. The wrap shall be secured in place with pressure sensitive tape. The two springs shall be wrapped with greaseproofed barrier material. The wrap shall be secured in place with pressure sensitive tape. The hardware and the packets containing the pins and springs shall be placed together and wrapped with greaseproofed barrier material. The wrapped items shall be overwrapped with paperboard cushioning material. The wrap shall be secured in place with gummed tape.

5.3.1.4.4 Boot and clamps.- The clamps shall be wrapped with greaseproofed barrier material. The wrap shall be secured in place with pressure-sensitive tape. The rubber boot shall be dusted lightly with commercial talcum. The boot and wrapped clamps shall be immobilized in a fiberboard box, conforming to type I or II, class 1, any style except FTC, TS, OPF and FPF, of Specification PPP-B-636, in such a manner that the boot is not deformed when the box is closed. The box shall be closed with two strips of 2 inch wide gummed tape. The tape shall be applied over the top and bottom seams and extend the full depth of the box onto each end panel.

5.3.1.4.5 Plywood base.- A plywood base shall be fabricated as shown in Figures 1 and 2.

5.3.1.4.6 Water-vaporproof barrier securing cam and transmitter to the base securing packet to cam, and securing fiberboard box to transmitter.- Water-vaporproof barrier material, conforming to class 1 of Specification MIL-B-131, shall be used in fabricating a floating bag. The barrier material shall contain appropriate size bolt holes for mounting bolts to pass through. Gaskets with appropriate size bolt holes to permit the mounting bolts to pass through shall be cemented to each side of the water-vaporproof barrier in such a manner that the holes in the gaskets and the holes in the barrier coincide. The gaskets shall be fabricated from material not less than 3/32 inch thick and conforming to class II grade 60 of Specification MIL-R-6855. The adhesive shall conform to type III of Specification MIL-A-5092. The aforementioned adhesive shall be applied on the surfaces of the gaskets which will contact the plywood base when the barrier is placed on the base.

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The barrier shall then be positioned on the base in such a manner that the bolt holes in the base, the barrier and gaskets coincide. Greaseproofed barrier material, containing bolt holes for bolts to pass through for securing the cam to the base, shall be positioned on the gasket material. The size of the greaseproofed barrier material shall be such that it will cover the cam completely after the cam is bolted in place. The cam shall be positioned on the greaseproofed barrier material and secured in place with appropriate size step bolts, washers, lock washers and nuts shown in Figure 1. The cam shall then be covered with the greaseproofed barrier material, which shall be secured in place with pressure-sensitive tape. The packet containing the cam locating pins, springs, and mounting hardware, specified in 5.3.1.4.3 shall be secured to the greaseproofed covered cam with waterproof pressure sensitive tape. The plywood pad, shown in Figure 1, shall be positioned on the gasket material. A 6 by 6 inch piece of greaseproofed barrier material, containing 4 holes for screws to pass through for securing the transmitter to the pad and base, shall be positioned on the pad. The transmitter shall be positioned on the greaseproofed barrier material and secured in place with the hexagon head bolts and washers shown in Figure 1. The rubber pads and metal clamp, shown in Figure 1, and the arm of the transmitter shall be positioned as shown in Figure 5. The metal clamp shall be secured to the base with step bolts, washers, lock washers and nuts shown in Figure 1. The box containing the boot and clamps, specified in 5.3.1.4.4 shall be secured to the housing of the transmitter with waterproof pressure-sensitive tape as shown in Figure 5. Unthinned paint shall be applied on the exposed threads above the nuts on the bolts used to secure the items to the plywood base.

5.3.1.4.7 Water-vaporproof bag.- After the items are secured in place on the water-vaporproof barrier covered base, the water-vaporproof barrier shall be heat sealed in such a manner as to form a bag. Entrapped air shall be exhausted prior to final closure. The water-vaporproof barrier shall be heat sealed in such a manner that the bag will pass the heat seal and quick leak tests of Specification MIL-P-116.

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5.3.2 Level C.

5.3.2.1 Cleaning and drying.- Cleaning and drying shall be as specified in 5.3.1.1 and 5.3.1.2.

5.3.2.2 Unit packaging.- The transmitter shall be packaged as specified in 5.3.1.4.1 to 5.3.1.4.6 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116.

(a) Paper material, conforming to type II of Specification MIL-P-130, shall be used in lieu of greaseproofed barrier material specified under Level A unit packaging.
(b) Talcum, specified in 5.3.1.4.4, shall not be dusted on the boot.
(c) The water-vaporproof barrier, gaskets and adhesive, specified in 5.3.1.4.6 shall be omitted.
(d) The size of the paper material which is placed between the cam and the plywood base in lieu of the greaseproofed barrier material shall be approximately equal to the size of the surface of the cam which contacts the barrier material.

5.4 Packing.

5.4.1 Level A.- Each plywood base, containing a packaged transmitter, shall be immobilized in a shipping container by nailing a wood hold-down cleat in each corner of the container. Each hold-down cleat shall be 2 by 3 inches and the length shall be such that the cleat will be flush with the top edge of the container. The rubber pads on the corners of the plywood base shall be compressed at least 1/8 inch when the hold-down cleats are nailed in place. Cement coated nails shall be used.The length of the nails and nailing pattern shall be as specified in the applicable box specification. The shipping container shall conform to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2, plywood
PPP-B-621 - Class 2 (oversea type) - style 2 or 4

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5.4.2 Level B or C.- One unit packaged transmitter shall be immobilized in a shipping container as specified in 5.4.1. The shipping container shall conform to one of the following specifications:

PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), style 2 or 4

5.4.3 Strapping.- Strapping shall be applied on each shipping container as specified in the applicable container specification.

5.5 Marking.

5.5.1 Interior packages.- Marking of interior packages and water-vaporproof bag shall be in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package.

5.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A or B shipping container and omitted on the Level C container. Fragile labels shall be affixed to the container in accordance with Standard MIL-STD-129.

5.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack.- Prior to quantity packaging production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

5.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedures specified herein, shall be in accordance with the highest grades of practice associated with this type of work.

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6. NOTES

6.1 Intended use.- The Transmitter Superelevation: M23 is a component unit of the optical electro-mechanical fire control system as used in the 120mm Gun Tank: M103A1. The function of the transmitter is to subtract the superelevation supplied by the computer from the gun elevation and to transmit the difference electrically to an elevation servomechanism of the tank fire control system.

6.2 Ordering data.- Procuring documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery. (See 3.1)

6.3 Size and weight of packed shipping container.- The following inside dimensions, overall cubage and gross weight are approximate for one unit packaged transmitter packed in a nailed wood box for Level A:

Inside dimensions	- 30.5 inches long by 13.5 inches wide by 12 inches deep
Cubage	- 3.8 cubic feet
Gross weight	- 92.5 pounds

6.4 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.5 Definitions.

6.5.1 Tactile.- Skillful manipulation or sensing.

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6.5.2 Special testing device(DST).- Special testing devices are defined as special inspection fixtures instruments and modified standard measuring equipment.

6.5.3 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

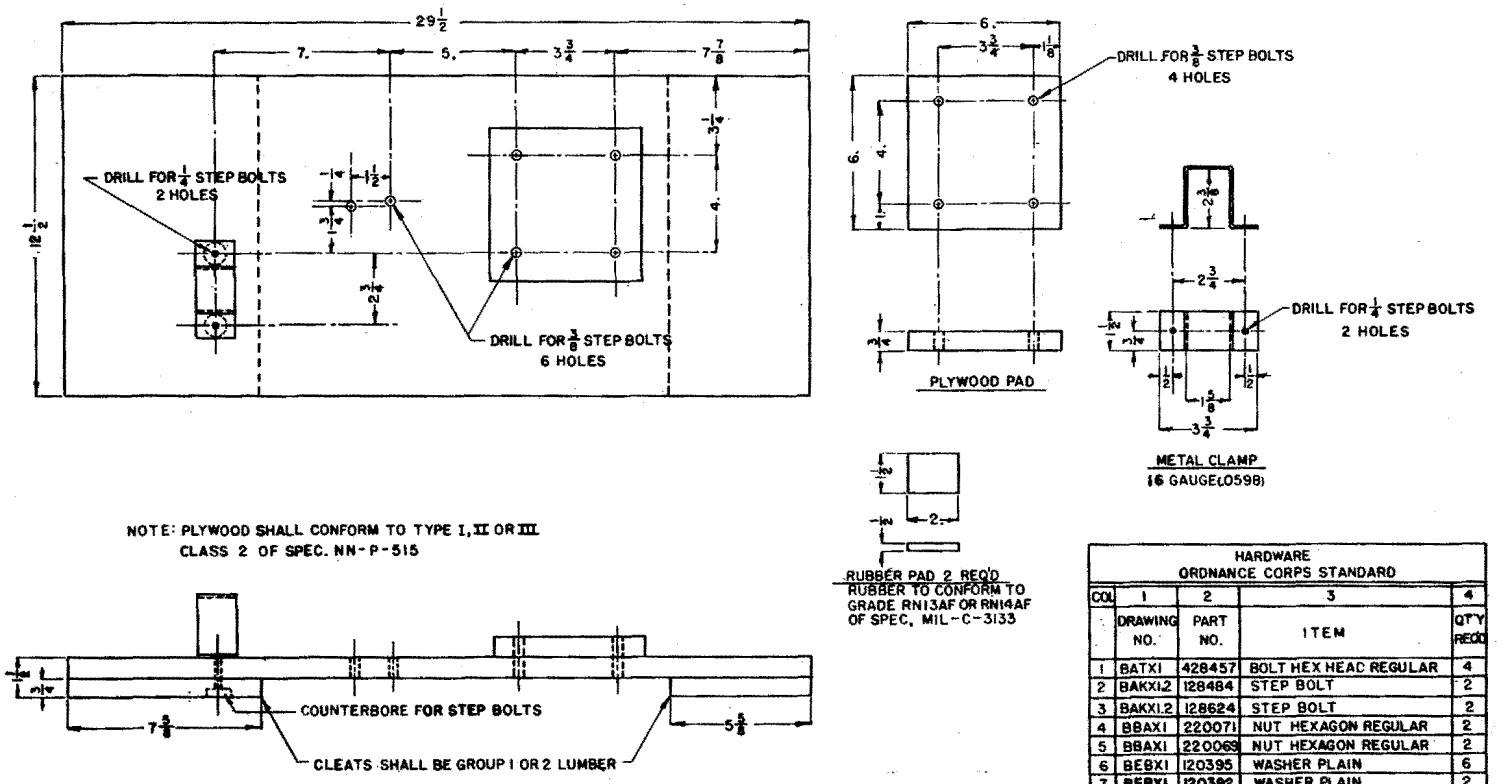
Custodian:

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Preparing Activity:

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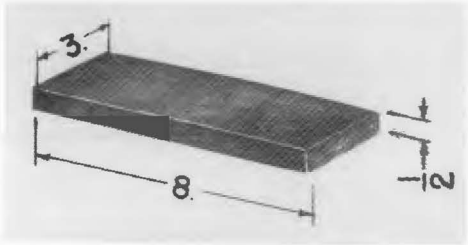


HARDWARE ORDNANCE CORPS STANDARD				
COL	1	2	3	4
	DRAWING NO.	PART NO.	ITEM	QT'Y REQD
1	BATX1	428457	BOLT HEX HEAD REGULAR	4
2	BAKX1.2	128484	STEP BOLT	2
3	BAKX1.2	128624	STEP BOLT	2
4	BBAX1	220071	NUT HEXAGON REGULAR	2
5	BBAX1	220069	NUT HEXAGON REGULAR	2
6	BEBX1	120395	WASHER PLAIN	6
7	BEBX1	120392	WASHER PLAIN	2
8	BECX2	121637	WASHER LOCK	2
9	BECX2	120381	WASHER LOCK	6

PLYWOOD BASE FOR SUPERELEVATION TRANSMITTER M23

FIGURE 1

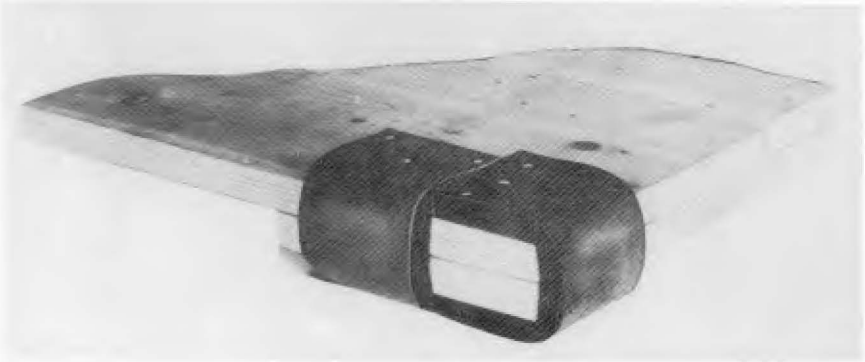
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RUBBER STRIP

NOTES: RUBBER STRIP SHALL CONFORM TO GRADE RN 13 AF OR RN 14 AF OF SPECIFICATION MIL-C-3133.

THE RUBBER SHALL BE COMPRESSED AT LEAST 1/8 INCH WHEN THE WOOD BASE IS IMMOBILIZED IN THE SHIPPING CONTAINER.



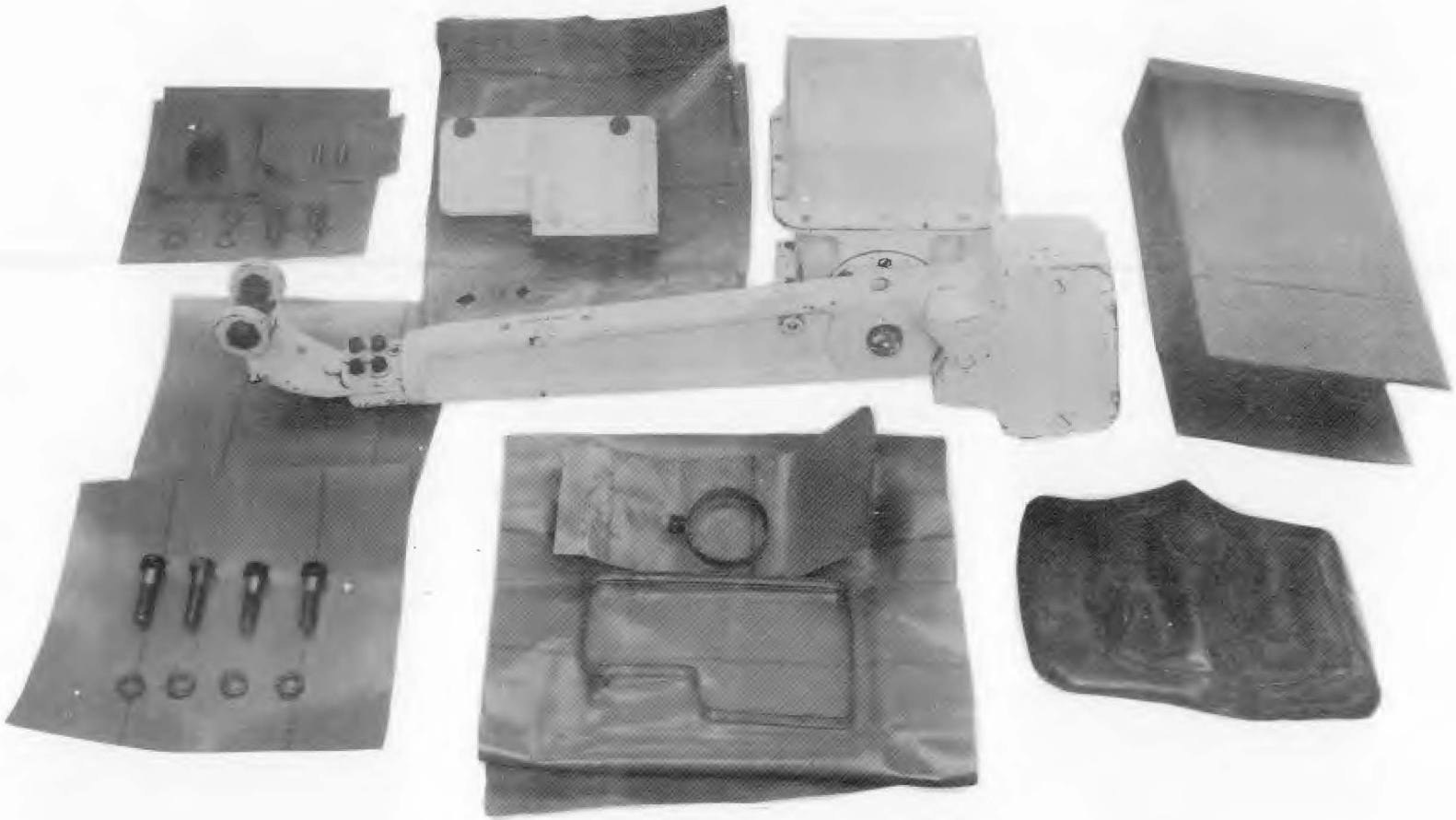
EACH END OF RUBBER STRIP SHALL BE SECURED TO BASE WITH NAILS. (DRG. BPAX 3, PART NO. 116892)

EIGHT RUBBER STRIPS ARE REQIURED FOR CORNERS OF BASE. IF STRIPS ARE ALSO APPLIED AT CENTER OF BASE, 10 ARE REQIURED.

VIEW SHOWING RUBBER STRIPS SECURED TO CORNER OF BASE

FIGURE 2

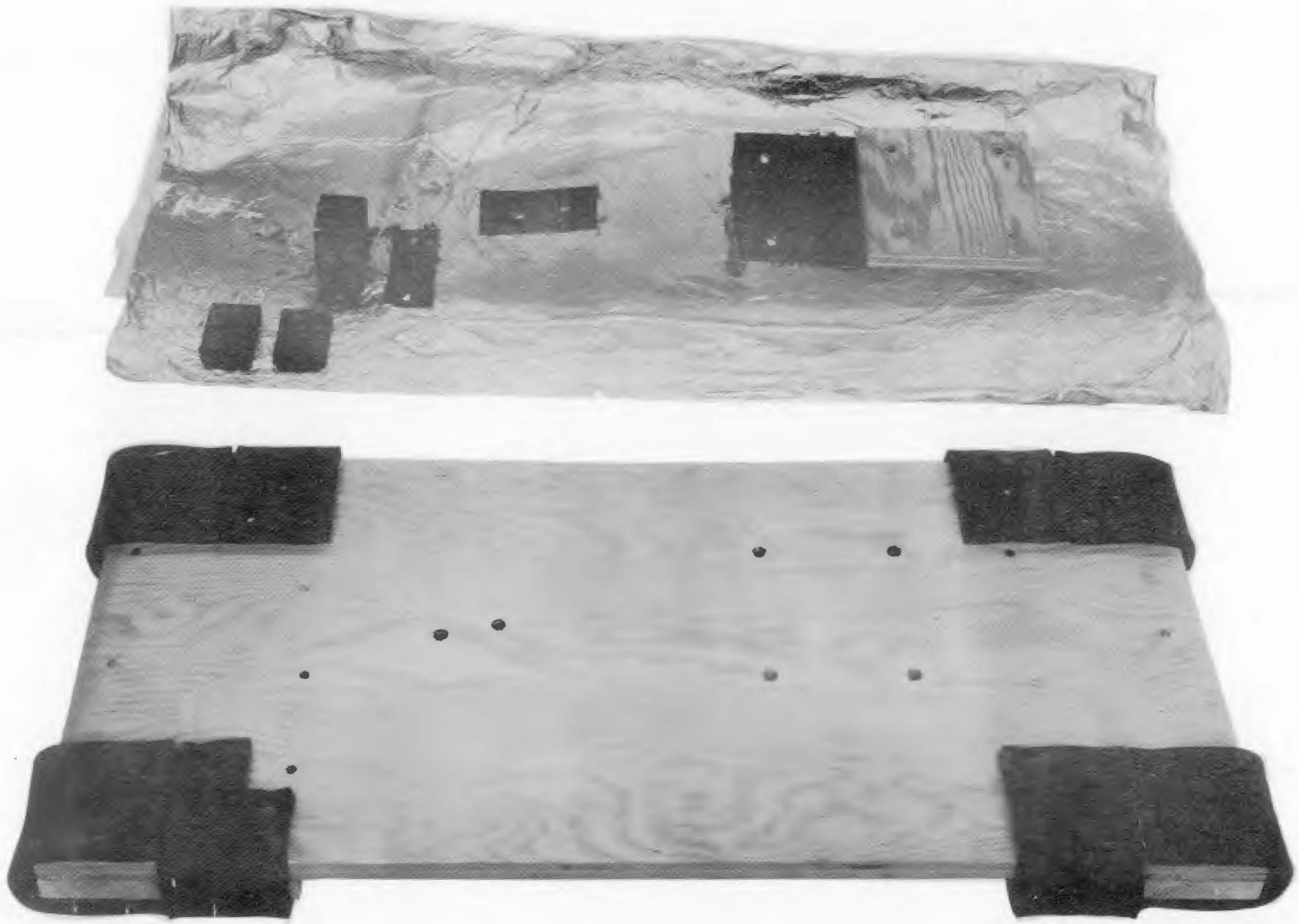
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SUPERELEVATION TRANSMITTER M23 **showing all related loose items prior to mounting on base**

FIGURE 3

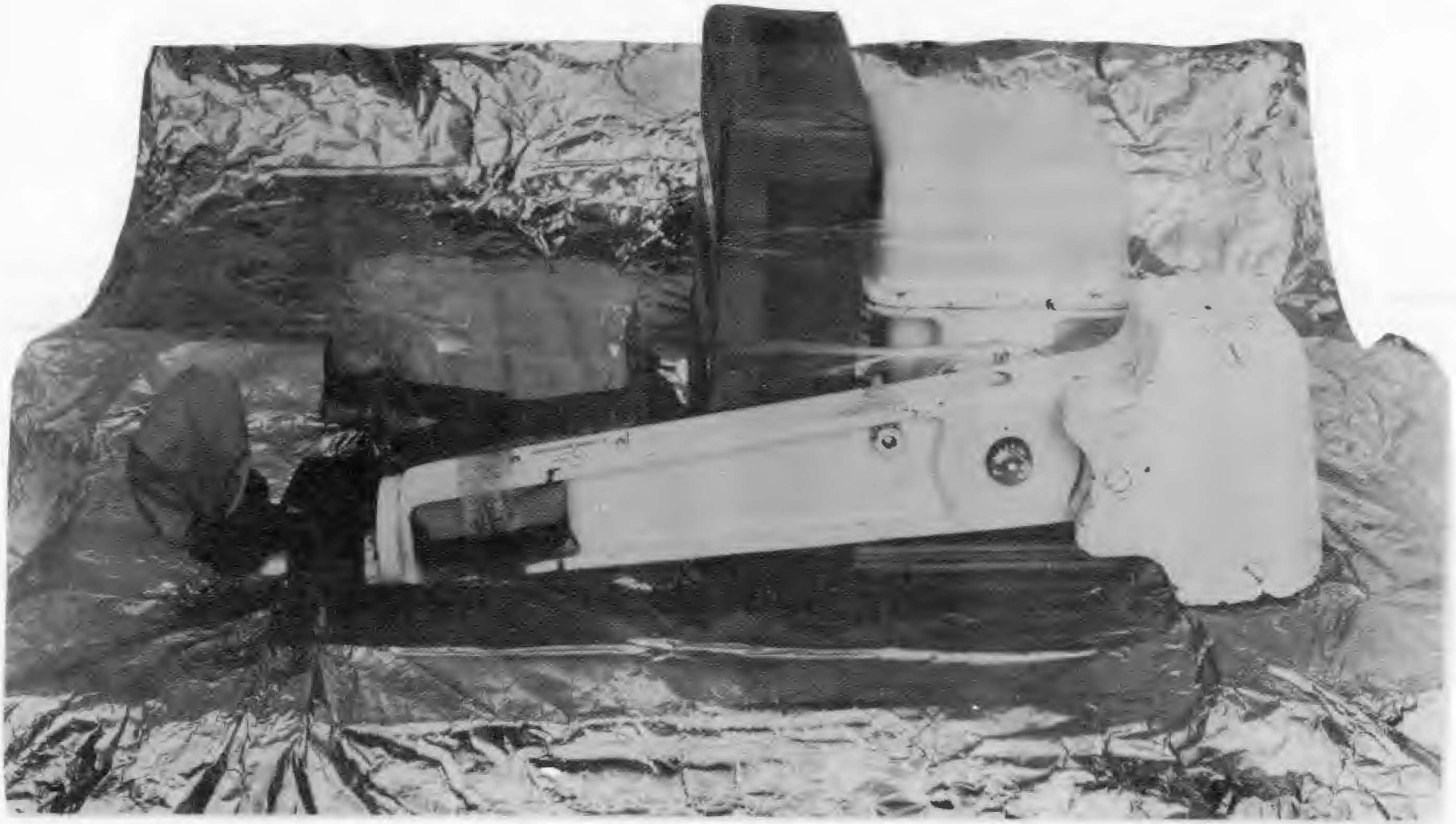
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SUPERELEVATION TRANSMITTER M23 showing plywood base and watervaporproof barrier

FIGURE 4

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SUPERELEVATION TRANSMITTER M23 mounted on base prior to sealing watervaporproof barrier

FIGURE 5

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